

Project Vault: Investment Memorandum

Structured Asset Comparison & Capital Budgeting

Sensitivity

Lead Proponent Group
Financed via Green Bond Framework Structuring

June 2026

1 Executive Summary & Strategic Taxonomy

This memorandum evaluates three discrete energy storage methodologies within the Finnish bidding zone of Nord Pool:

- Modular Micro-PSH (10 MW / 40 MWh):** Low-risk, behind-the-meter (BTM) industrial placement targeting localized wind-smoothing and day-ahead peak-shaving.
- Strategic Large PSH (200 MW / 1,600 MWh):** Sub-surface brownfield mining infrastructure utility asset (e.g., deep-shaft gravity storage) co-optimized across reserve capacity and wholesale arbitrage.
- Utility-Scale BESS (50 MW / 100 MWh):** Lithium-Iron-Phosphate (LFP) rapid-response containerized battery asset dedicated to fast frequency stabilization.

2 Detailed Revenue Mechanics by Asset Class

2.1 Strategy 1: Modular Micro-PSH (10 MW / 40 MWh)

Operates primarily via localized industrial cost-avoidance and daily spot price arbitrage.

- Arbitrage Mechanics:** Charges during low-cost or negative price intervals on Nord Pool (capturing surplus wind blocks below 10 EUR/MWh) and discharges during daily peak consumption intervals.
- Calculation Baseline:** Assumes 1.5 cycles per day (547 cycles per annum) with an average captured gross spread of 85 EUR/MWh and a round-trip efficiency (η) of 76%.
- Annual Revenue Formula:**

$$R_{\text{micro}} = 40 \text{ MWh} \times 547 \text{ cycles} \times 85 \text{ EUR/MWh} \times 0.76 (\eta) = 1,417,616 \text{ EUR}$$

2.2 Strategy 2: Strategic Large PSH (200 MW / 1,600 MWh)

Dual-optimization across Fingrid reserve procurement and large wholesale Nord Pool spreads.

- Fingrid Capacity Reserves (100 MW Allocation):** Committed to automatic Frequency Restoration Reserves (aFRR) and Fast Frequency Containment (FCR-D). At a conservative clearing price of 25 EUR/MW/h and full theoretical availability:

$$R_{\text{reserve, nominal}} = 100 \text{ MW} \times 8,760 \text{ hours} \times 25 \text{ EUR/MW/h} = 21,900,000 \text{ EUR}$$

Accounting for an operational availability factor ($AF_{\text{res}} = 90\%$) for planned maintenance, baseline revenue yields 19,710,000 EUR.

- **Wholesale Arbitrage (100 MW Allocation):** Deep discharge (800 MWh blocks) targeting 5 cycles per week (260 cycles per annum). Captures structural spreads driven by localized wind drop-offs or nuclear outages (*Kireä pakkaspäivä* cycles), averaging a 100 EUR/MWh spread at 78% round-trip efficiency (η).

$$R_{\text{arbitrage}} = 800 \text{ MWh} \times 260 \text{ cycles} \times 100 \text{ EUR/MWh} \times 0.78 (\eta) = 16,224,000 \text{ EUR}$$

- **Total Base Revenue:** 38,124,000 EUR per annum (nominal) / 35,934,000 EUR (availability-adjusted).

2.3 Strategy 3: Utility BESS (50 MW / 100 MWh)

High-frequency ancillary service play targeting fast-acting, short-duration capacity markets.

- **FCR-D Market Optimization:** 50 MW committed to FCR-D up/down regulation markets at an average market clearing yield of 30 EUR/MW/h.

$$R_{\text{bess, nominal}} = 50 \text{ MW} \times 8,760 \text{ hours} \times 30 \text{ EUR/MW/h} = 13,140,000 \text{ EUR}$$

Adjusted for 92% grid availability factor, baseline operational yield clears at 12,088,800 EUR.

- **Degradation Factor & Overhaul Cycle:** Electrochemical throughput incurs an active 2.5% annual capacity degradation. The asset requires a major cell stack repowering (40% of initial CapEx) every 12 years to reset performance.

3 Comparative Capital Budgeting Matrix

Table 1: Granular Cost and Lifespan Profiles

Parameter	Modular Micro-PSH	Strategic Large PSH	Utility BESS
Target Capacity Block	10 MW / 40 MWh	200 MW / 1,600 MWh	50 MW / 100 MWh
Base EPC Civil Cost	EUR 8,500,000	EUR 118,000,000	EUR 4,000,000
Electro-Mechanical / Cells	EUR 6,000,000	EUR 94,500,000	EUR 24,000,000
Grid Substation Hookup	EUR 1,500,000	EUR 40,500,000	EUR 4,000,000
Development & Permitting	EUR 1,000,000	EUR 27,000,000	EUR 1,000,000
Total Initial CapEx	EUR 17,000,000	EUR 280,000,000	EUR 33,000,000
Fixed Annual OpEx	EUR 306,000	EUR 5,040,000	EUR 660,000
Asset Economic Life	40 Years	80+ Years	12 Years (Repowered)

4 Sensitivity Analysis & Risk Underwriting

To account for power system evolution over time, we stress-test the annual base models against two key variables:

1. **Volatility Factor (σ_{spot}):** Scaling spot market price deviations (1.0 = current market state; 1.4 = wind-distorted spreads; 0.7 = contracted spread compression).
2. **Saturation Coefficient (α_{sat}):** Downward scaling of Fingrid reserve pricing as battery capacity enters (1.0 = premium yield; 0.5 = moderate saturation; 0.4 = structural equilibrium floor).

Table 2: Annualized Net Operational Cash Flow Sensitivities (EUR Millions)

Scenario	Variable State	Modular Micro-PSH	Strategic Large PSH	Utility BESS
Base Case	$\sigma_{\text{spot}} = 1.0, \alpha_{\text{sat}} = 1.0$	1.11	33.08	12.48
High Volatility	$\sigma_{\text{spot}} = 1.4, \alpha_{\text{sat}} = 1.0$	1.68	39.57	12.48
Market Saturation	$\sigma_{\text{spot}} = 1.0, \alpha_{\text{sat}} = 0.5$	1.11	22.13	5.91
Severe Stress	$\sigma_{\text{spot}} = 0.7, \alpha_{\text{sat}} = 0.4$	0.69	15.08	4.60

5 Project Realization Timeline (2026–2031)

Table 3: Comparative Deployment Milestones

Phase	Modular Micro-PSH	Strategic Large PSH	Utility BESS
Feasibility & Environmental (YVA)	Months 1–2	Months 1–18	Months 1–2
Permitting & Grid Interconnection	Months 3–4	Months 19–36	Months 3–4
Procurement & EPC Contracting	Months 5–6	Months 37–48	Months 5–6
Civil Works & Shaft Excavation	Months 7–8	Months 49–66	Months 7–10
Commercial Operation (COD)	Month 9	Month 68	Month 11

6 Dynamic Valuation & Multi-Decadal Capital Budgeting

To evaluate these investments under dynamic market conditions, we model time-dependent driver paths:

$$\alpha_{\text{sat}}(t) = 0.40 + \frac{0.60}{1 + e^{0.4(t-3)}}, \quad \sigma_{\text{spot}}(t) = 1.0 + \ln(1 + 0.15 \cdot t)$$

Large PSH construction outlays are modeled across an S-curve disbursement schedule throughout Years 1–5 (10%, 20%, 30%, 25%, 15%). Cash flows are evaluated at a 5% corporate discount rate.

Table 4: Asset Valuation Summary (5% Corporate Discount Rate)

Financial Metric	Modular Micro-PSH	Strategic Large PSH	Utility BESS
Initial Capital Outlay	EUR 17.0M	EUR 280.0M (Phased)	EUR 33.0M
Commercial Operation	Year 2	Year 6	Year 2
Asset Economic Lifespan	40 Years	80+ Years	12 Years (Repowered)
Standalone Lifecycle IRR	12.99% (40-Yr)	10.91% (50-Yr)	13.55% (12-Yr)
Standalone Lifecycle NPV	EUR 26.9M	EUR 327.4M	EUR 12.9M
50-Year Multi-Gen NPV	EUR 26.9M	EUR 327.4M	EUR 31.2M
Accounting Break-Even	Year 10	Year 14	Year 6

6.1 Underwriting Synthesis & Strategic Conclusion

The structured capital comparison reveals two divergent risk-return archetypes:

- **Utility BESS** functions as a rapid-payback yield vehicle (Break-Even Year 6; 12-Year IRR 13.55%). However, because it lacks duration depth, it is acutely vulnerable to Fingrid reserve price decay ($\alpha_{\text{sat}} \rightarrow 0.40$) and incurs cyclic reinvestment liabilities every 12 years.
- **Strategic Large PSH** requires deep institutional financing during its 5-year civil excavation period, breaking even in Year 14. Once commissioned, however, its 8-hour storage depth allows it to pivot seamlessly from stabilizing reserves into booming day-ahead wholesale spreads ($\sigma_{\text{spot}} \uparrow$), yielding EUR 327.4M in NPV at an institutional-grade 10.91% IRR across a multi-generational horizon.